

Patch/Vulnerability Report Builder

Problem Statement

Vulnerability scanners generate reports containing numerous CVEs and technical security details that are difficult for managers and non-technical stakeholders to understand. Security teams spend significant time manually analyzing vulnerabilities, prioritizing risks, and preparing reports for decision-makers.

This manual process can delay remediation efforts and increase the risk of overlooking critical security issues.

Solution

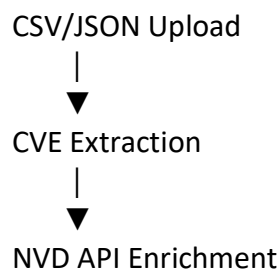
Patch/Vulnerability Report Builder is an AI-powered cybersecurity reporting platform that converts raw vulnerability scan exports into executive-friendly security reports.

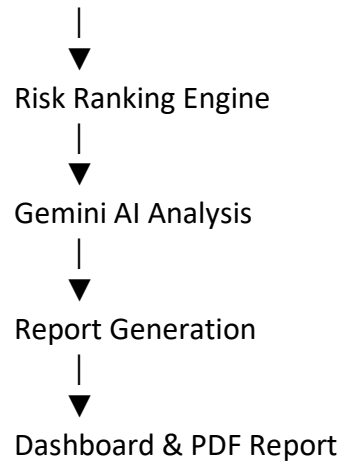
The system processes CSV or JSON scan files, enriches CVE information using the NVD API, ranks vulnerabilities based on risk, and uses Gemini AI to generate business impact assessments, remediation recommendations, and executive summaries.

Workflow

1. Upload vulnerability scan file (CSV/JSON)
2. Extract and validate CVE IDs
3. Retrieve vulnerability details from NVD API
4. Rank vulnerabilities using CVSS scores
5. Generate AI-powered insights using Gemini
6. Display results on dashboard
7. Generate downloadable Markdown and PDF reports

Architecture





Technology Stack

- Python
- Streamlit
- Gemini 2.5 Flash
- NVD API
- Pandas
- Plotly
- Pytest

Outcome

The solution transforms complex vulnerability data into actionable security intelligence, enabling organizations to quickly identify critical risks, prioritize remediation efforts, and communicate security findings effectively to stakeholders.